

TARTU, Estonia

WMO: 262420

Lat: 58.2642N

Lon: 26.4614E

Elev: 236

StdP: 14.57

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.4	-0.7	-11.6	3.0	-6.8	-5.4	4.1	0.0	21.4	29.9	19.7	31.0	3.7	80	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	82.1	67.2	78.8	65.3	75.6	63.7	69.4	78.2	67.2	75.6	65.3	73.2	7.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.3	73.8	64.2	90.8	71.4	62.2	84.4	69.3	33.6	78.0	31.8	75.6	30.3	73.2	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	17.0	15.2	DB	-13.0	87.0	8.2	2.8	-18.9	89.0	-23.7	90.6	-28.3	92.2	-34.3	94.2	(4)
(5)			WB	-13.4	72.7	8.1	2.2	-19.2	74.2	-24.0	75.5	-28.5	76.7	-34.4	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	23.1	22.8	30.5	42.7	52.1	59.3	64.5	62.1	53.6	43.1	34.9	28.1		
	DBStd	16.64	11.50	11.99	8.27	7.49	7.73	6.01	5.63	5.38	6.06	7.21	7.40	9.86		
	HDD50	3908	833	760	606	243	66	3	0	0	34	231	452	680		
	HDD65	8084	1298	1180	1071	670	406	191	77	123	343	678	902	1145		
	CDD50	1422	0	0	0	23	130	282	451	376	142	18	0	0		
	CDD65	122	0	0	0	0	5	19	63	34	1	0	0	0		
	CDH74	1015	0	0	0	2	58	170	517	258	10	0	0	0		
	CDH80	178	0	0	0	0	6	23	105	44	0	0	0	0		
	CDH80	178	0	0	0	0	6	23	105	44	0	0	0	0		
(14) Wind	WSAvg	7.1	7.6	7.1	7.6	7.3	6.8	6.6	6.0	6.0	6.6	7.2	7.9	8.3		
(15) Precipitation	PrecAvg	24.1	1.4	1.1	1.1	1.3	2.0	2.5	3.0	3.2	2.6	2.2	2.0	1.7		
	PrecMax	34.8	3.0	3.0	3.0	2.8	4.1	5.2	6.5	8.1	5.3	4.9	4.4	3.4		
	PrecMin	15.5	0.2	0.3	0.2	0.0	0.4	0.2	0.4	0.6	0.6	0.4	0.4	0.5		
	PrecStd	4.2	0.7	0.6	0.7	0.7	0.9	1.3	1.6	1.8	1.1	1.0	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	43.0	55.8	71.8	80.7	82.8	86.5	84.5	75.4	62.2	51.5	48.0		
		MCWB	41.7	40.2	44.8	54.6	65.0	66.8	69.8	67.3	61.6	55.3	48.4	46.2		
	2%	DB	39.8	40.3	48.4	66.0	75.1	78.9	82.9	80.3	69.8	57.7	48.5	44.2		
		MCWB	38.5	38.2	41.2	51.6	60.4	64.5	68.3	66.3	59.8	53.6	46.3	42.6		
	5%	DB	37.5	38.3	43.8	61.0	70.9	75.1	79.8	76.7	66.3	55.3	46.6	42.0		
		MCWB	36.3	36.5	38.5	49.2	57.4	63.0	66.7	64.1	58.2	52.1	45.0	40.6		
	10%	DB	35.7	36.4	40.7	56.2	66.5	71.4	76.5	73.2	63.2	53.1	44.7	39.3		
		MCWB	34.7	35.0	36.8	46.6	55.1	60.5	65.2	62.6	56.7	50.4	43.3	38.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	40.6	45.9	56.6	67.2	69.6	72.7	71.1	63.9	56.7	49.0	46.1		
		MCDB	43.4	41.9	53.9	67.5	76.7	79.0	82.0	79.6	69.9	59.9	50.4	47.6		
	2%	WB	38.6	38.4	41.8	53.1	62.9	66.6	70.2	67.9	61.4	54.5	46.6	42.8		
		MCDB	39.4	39.7	46.7	63.2	71.4	75.7	79.4	75.7	67.4	57.0	47.8	43.8		
	5%	WB	36.4	36.7	39.3	50.2	59.5	64.2	68.2	65.7	59.4	52.7	45.1	40.5		
		MCDB	37.2	38.0	42.7	58.9	67.6	72.4	76.8	73.5	64.6	55.1	46.3	41.6		
	10%	WB	34.7	35.0	37.4	47.5	56.6	62.0	66.3	63.9	57.6	50.7	43.2	38.1		
		MCDB	35.5	36.2	39.9	55.1	64.8	69.5	74.3	71.4	61.8	52.8	44.2	39.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	10.2	12.8	17.1	18.8	17.0	17.1	17.1	14.7	10.1	6.9	7.1		
		MCDBR	6.5	7.7	16.2	25.2	24.6	21.9	21.9	22.1	18.7	11.9	8.1	6.6		
	5% WB	MCWBR	6.3	6.7	10.5	13.5	12.7	10.9	10.1	10.3	10.2	8.6	7.2	6.4		
		MCWBR	6.5	7.1	14.0	21.8	21.3	19.3	19.6	19.2	16.0	10.9	7.5	6.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.274	0.305	0.335	0.368	0.363	0.362	0.387	0.384	0.354	0.334	0.312	0.267		
		taud	2.123	2.137	2.195	2.320	2.395	2.424	2.374	2.379	2.449	2.401	2.219	2.111		
	Ebn at Noon	Ebn at Noon	173	224	254	266	276	278	267	259	249	218	157	135		
		Edh at Noon	16	26	32	34	34	33	35	32	26	20	15	12		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	114	324	783	1233	1676	1715	1656	1309	835	368	115	69		
	RadStd	RadStd	22	58	98	148	179	142	167	133	98	55	16	12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	-293	-297	N/A	N/A

Nomenclature: See separate page